

HOMEWORK 11

Due date: Monday of Week 12

The following are exercises from M. Artin's book "Algebra", 2nd edition.

Exercises: 2.4, 2.6, 3.1, 3.2, 4.1, 4.2, 4.3, 4.4, 4.5, 4.6, 4.8, 5.1, 5.2, 5.3, 5.6, 6.2, 6.3, 6.5, 6.6, 6.7, 6.10, 6.11, 9.4, 9.6, pages 69-75

For Exercises in Section 9, just remind you the notations. $2x \equiv 5$ modulo 9 just means that $2x = 5$ in $\mathbb{Z}/9\mathbb{Z}$. For the Chinese Remainder Theorem of Ex.9.6, recall that we have proved the following. Let F be a field, and $f, g \in F[x]$ be relative prime polynomials, then

$$F[x]/(fg) \cong F[x]/(f) \times F[x]/(g).$$

Try to interpret Ex.9.6 using a similar isomorphism as above. You can also try to prove that isomorphism.

Problem 1. Let K be a field and $F \subset K$ is a subfield. Let V, W be two vector spaces over F and let $T : V \rightarrow W$ be an F -linear map. Consider the T -linear map $T \otimes \text{id} : V \otimes_F K \rightarrow W \otimes_F K$. Show that

$$\text{Ker}(T \otimes \text{id}) = \text{Ker}(T) \otimes_F K.$$

Problem 2. Let K be a field and $F \subset K$ is a subfield. Let V be a vector space over F . Let $S = \{\alpha_1, \dots, \alpha_n\} \subset V$ be a subset of V and consider $S' = \{\alpha_1 \otimes 1, \dots, \alpha_n \otimes 1\} \subset V \otimes_F K$. Show that S is linearly independent over F if and only if S' is linearly independent over K .

Problem 3. Let $G = \text{SL}_2(\mathbb{Z})$. In other words, $G = \{g \in \text{GL}_2(\mathbb{Z}) \mid \det(g) = 1\}$. Consider the following elements

$$U = \begin{bmatrix} 1 & 1 \\ & 1 \end{bmatrix}, \quad L = \begin{bmatrix} 1 & \\ & 1 \end{bmatrix}, \quad T = \begin{bmatrix} & 1 \\ -1 & \end{bmatrix}$$

of G .

- (1) Is it true that G is generated by U and L ?
- (2) Is it true that G is generated by U and T ?
- (3) Write T as a product of U , L and their inverses.
- (4) Consider the matrices

$$A = \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix} \in G.$$

How to write A as a product of U, L and their inverses? How to write A as a product of U and T as their inverses?

- (5) Is it possible to write $-I$ as a product of U and T and their inverses?

Problem 4. (1) Let $n \geq 3$. Show that $Z(S_n) = \{1\}$, where $Z(S_n)$ denotes the center of S_n .
 (2) Show that $Z(\text{GL}_n(F)) = \{aI_n \mid a \in F^\times\}$.

Let G be a group. For $x, y \in G$. Recall that the commutator of x, y is defined to be $[x, y] = xyx^{-1}y^{-1}$. Let $[G, G]$ be the subgroup of G which is generated by all commutators $[x, y]$ for all $x, y \in G$.

Problem 5. Let F be a field and $G = \text{GL}_n(F)$, where $n \geq 2$. Show that $[G, G] = \text{SL}_n(F)$.

Hint: It is easy to show $[G, G] \subset \text{SL}_n(F)$: just think about what is $\det([G, G])$? The other direction is harder. To show $\text{SL}_n(F) \subset [G, G]$, one need to find a set of generators of $\text{SL}_n(F)$, (similar to Problem 3), and show each generator can be written as a product of commutators. See Hoffman-Kunze, Exercise 16, page 107, for a similar problem.

Similar as the above one, one can show that $[S_n, S_n] = A_n$ if $n \geq 5$. We will talk this later.

Let G be a group. Recall that $\text{Aut}(G) = \{f : G \rightarrow G \mid f \text{ is an isomorphism}\}$. We have defined a homomorphism

$$\varphi : G \rightarrow \text{Aut}(G)$$

defined by

$$g \mapsto \varphi_g,$$

where $\varphi_g \in \text{Aut}(G)$ is defined by $\varphi_g(x) = gxg^{-1}$. We know the kernel of φ is $Z(G)$, the center of G . The image of φ is $\text{Inn}(G)$, the group of inner automorphisms of G .

Theorem 0.1 (Automorphisms of S_n). (1) *If $n \neq 2, 6$, then $\varphi : S_n \rightarrow \text{Aut}(S_n)$ is an isomorphism. In particular, we have $\text{Aut}(S_n) = \text{Inn}(S_n) \cong S_n$ if $n \neq 2, 6$.*

(2) *If $n = 2$, then $\varphi : S_2 \rightarrow \text{Aut}(S_2)$ is surjective but not injective. In this case, $\text{Aut}(S_2) = \{1\}$.*

(3) *If $n = 6$, then $\varphi : S_6 \rightarrow \text{Aut}(S_6)$ is injective but not surjective.*

We won't prove this. See [this wikipedia link](#). The outer automorphisms (elements in $\text{Aut}(G)$ but not in $\text{Inn}(G)$) of S_6 are interesting. See the above link for the constructions. It is also related to icosahedron, see page 183 of Artin's book.